Hypertension Monitoring Platform — Project Spec

Oct 30, 2026 · @Rey

Overview

We turn the months of silence between two consultations into a continuous record of home blood pressure readings, and show the doctor only the patients who need attention today.

The problem. About 7.7 million people in Algeria have hypertension, around 6.4 million of them uncontrolled (figures from the hackathon brief). Doctors see one reading per visit, so a rising trend or a crisis between visits goes unnoticed.

The solution. A patient mobile app to record readings and symptoms, an SMS and pharmacy channel for patients without a smartphone, and a doctor dashboard that sorts patients by risk, explains every alert and tracks measurable indicators.

Principle. The system assists; the doctor decides. It never diagnoses or changes a treatment.

Pitch line: "The AI explains, the guidelines decide, and the doctor stays in charge."

Working name: TensioCare (to be changed by the team).

Users and roles

Three roles share one backend; each sees only what it needs.

Role	Uses	Main job	Sees
Patient	Mobile app, or SMS	Record readings, symptoms and pills; follow advice	Own readings, colour status, advice, doctor messages
Doctor	Web dashboard	Review prioritised alerts, follow patients, prepare consultations	Own patients only, charts, alerts, indicators

Role	Uses	Main job	Sees
Pharmacist or nurse (optional)	Simple web form	Measure and enter readings for patients without a device; screening campaigns	Entry form and confirmation only, no history

A patient is linked to one doctor through an invite code the doctor gives at the consultation.

Patient mobile app

The app must be usable by an elderly patient in under 30 seconds per reading, in Arabic, Darija or French.

Screens

1. **Onboarding:** language choice, phone number with OTP, doctor invite code, basic profile (age, sex, treatment names, emergency contact).
2. **Home:** today's status as a colour (green, orange, red), next reminder, big "Add reading" button, last 7 days mini chart.
3. **Guided measurement:** a short checklist before measuring (sit 5 minutes, back supported, arm at heart level, no talking, 2 readings 1 minute apart).
4. **Add reading:**
   - Type systolic, diastolic and pulse, or
   - Take a photo of the monitor screen; on-device OCR fills the numbers, the patient confirms
5. **Context tags (one tap each):** took my pill / forgot my pill, salty meal, stress, poor sleep.
6. **Symptoms:** headache, dizziness, nosebleed, chest pain, vision problems, confusion, weakness on one side, difficulty speaking.
7. **Result screen:** colour, short explanation and the advice for this situation (see Alert system).
8. **History:** chart with the target zone, morning vs evening, filters by week and month.
9. **Advice chat:** questions answered only from approved content, in the patient's language.
10. **Contact my doctor:** call or message with the latest readings attached; outside the doctor's hours it shows "If urgent, call 14".

Key behaviours

- Works offline: readings are stored locally and synced when a connection returns; if no internet for a confirmed red reading, the app sends it by SMS automatically.
- Adaptive reminders (see Intelligence): fewer when stable, more when something changes.
- Large text and icons, minimal typing, voice-over friendly.

Doctor dashboard

The doctor opens the dashboard and sees, in one glance, which patients need attention today and why.

Screens

1. **Triage list (home):** all patients sorted by risk level (red, orange, yellow, green). Each row shows the last reading, 7-day home average, trend arrow, adherence %, and the reason for the flag.
2. **Alert inbox:** open alerts with their explanation ("4 of the last 6 readings above 135/85, slope +2.1 mmHg per week"). Actions: acknowledge, message the patient, schedule a visit, close.
3. **Patient page:**
   - Chart of systolic and diastolic over time with the target zone shaded
   - Morning vs evening split
   - Medication changes and symptoms marked on the timeline
   - Context patterns ("+15 mmHg on days with a missed pill")
   - Doctor notes and personal target thresholds per patient
4. **Pre-consultation report:** one page with the 7-day home average, adherence, trend, symptoms and alerts since the last visit; printable or PDF.
5. **Messages:** short replies to patients ("Come tomorrow", "Take your evening dose").
6. **Indicators:** see the Indicators section.
7. **Settings:** availability hours, alert thresholds per patient, daily digest time.

Workload rules

- Only red alerts notify in real time; everything else goes to one daily digest.
- Every alert says why it fired, so the doctor never has to open raw data to understand it.

- The doctor can mute or adjust thresholds for a specific patient.

Alert system

An abnormal reading is first confirmed by a second measurement, then classified into one of five levels; only the two red levels notify the doctor instantly.

Step 1, confirm. If a reading is out of range, the app asks: "Sit calmly for 5 minutes and measure again." The alert level uses the confirmed reading. This removes most false alarms (movement, talking, stairs).

Step 2, classify.

Level	Trigger (to validate with a doctor)	Patient sees	Doctor gets
Red, emergency	180/120 or higher with a danger symptom (chest pain, vision problems, confusion, weakness on one side, difficulty speaking)	"Call 14 or go to hospital now" with a one-tap call button	Immediate SMS and push
Red, crisis	180/120 or higher confirmed, no danger symptom	Rest, re-measure in 30 minutes, contact your doctor today	Immediate alert
Orange	160/100 or higher, or below 90/60 with dizziness or fatigue	Advice for the situation; "your doctor has been informed"	Alert at the top of the triage list
Yellow	Rising trend over 14 days, or home average above 135/85	Encouragement, reminder about pills and salt	Daily digest
Green	Within target	"Good, keep it up"	Nothing; chart only

Step 3, escalate if needed. A red alert not acknowledged by the doctor within a set time (for example 30 minutes) is re-sent, and the patient is reminded of 14.

Channels. Push notification when online; SMS fallback for both patient and doctor when not.

Intelligence

The intelligence is explainable rules and simple statistics, not a trained LLM: it works with 10 to 15 readings per patient and can always say why it raised an alert.

Three levels of analysis

Level	Method	Example output
Rules	Thresholds for home readings (hypertension at 135/85 or higher at home; crisis at 180/120)	"185/122 confirmed: crisis level"
Trends	7-day moving average and linear regression slope over 14 days; projection of the slope forward	"Systolic rising +2 mmHg per week; likely above target in about 10 days"
Patterns	Compare average readings with and without each context tag	"Readings 15 mmHg higher on days with a missed pill"

The trend projection is the "prediction" feature: honest, explainable and accurate enough for a short horizon.

Adaptive measurement schedule

The app asks for readings only when they matter, so patients keep using it.

Patient status	Reminders
Stable and controlled	2 to 3 days per week, morning and evening
Rising trend or near target	Every day, morning and evening
New treatment or recent high reading	Morning and evening for 7 days, then re-evaluated
Symptom reported	"Measure now", then again in 30 minutes

Each session = 2 readings 1 minute apart; the app stores their average.

Safe AI advice

1. An approved advice library: short messages per situation (high with headache, low with dizziness, missed pill, salty meal), based on hypertension guidelines and reviewed by a doctor.
2. The rules engine picks the right message from level, symptoms and context.
3. An LLM only rephrases and explains that message in the patient's language and answers follow-up questions using only the approved content (retrieval-augmented generation).
4. Hard limits: never changes a dose, never diagnoses, always redirects danger signs to 14 or the doctor.
5. The LLM also writes a short summary for the doctor on each patient page, generated from the computed numbers.

Demo data

No real dataset is needed: generate synthetic patients with realistic profiles (stable, slowly rising, crisis, non-adherent, over-medicated with low readings) to show every alert type.

Low-connectivity access

Any patient can send a reading without a smartphone or internet, which covers elderly and rural patients.

Channel	Needs smartphone	Needs internet	How
Mobile app	Yes	No (syncs later)	Normal flow; red readings fall back to SMS automatically
SMS	No	No	Patient sends `TA 145 92 78` (systolic, diastolic, pulse); optional symptom codes like `TA 182 118 80 CP` for chest pain
Pharmacy or nurse	No	Pharmacy side only	Professional measures and enters the reading with the patient's phone number

SMS replies. Every SMS reading gets an automatic reply with the colour and one line of advice, for example: "Tension elevee (162/98). Reposez-vous et remesurez ce soir. Votre medecin est informe."

Hackathon setup. An Android phone running an SMS gateway app forwards incoming SMS to the backend and sends replies. In production this becomes an operator short number.

Early detection mode. Pharmacies or screening campaigns measure anyone; a high reading triggers an SMS inviting the person to see a doctor, and the case is counted in the indicators.

Indicators

The brief explicitly asks for measurable indicators, so the doctor dashboard has a page that proves impact with numbers.

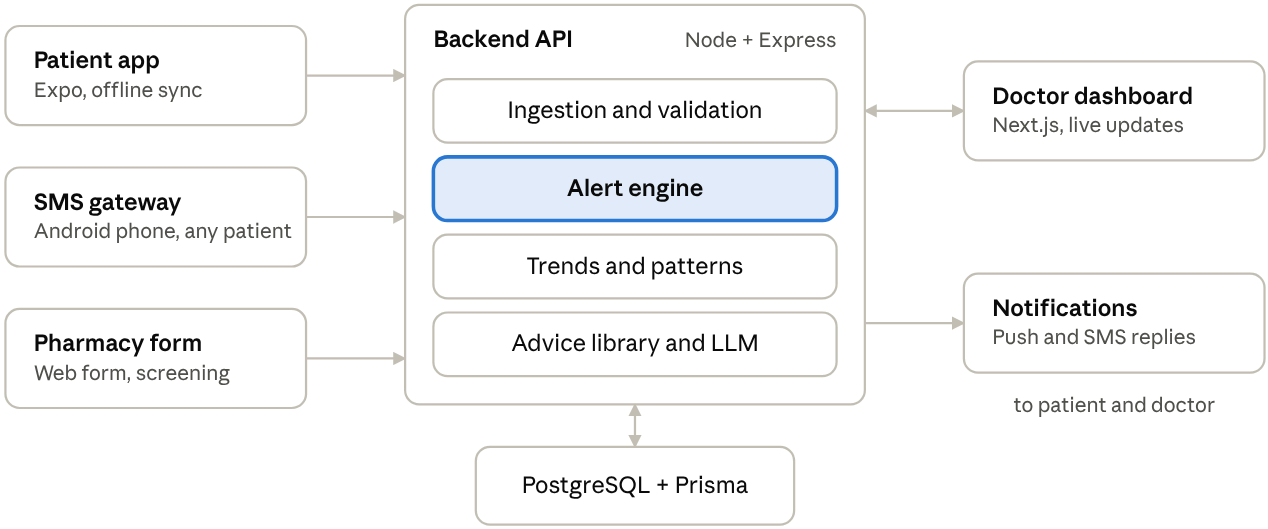
Indicator	Definition	Why it matters
Measurement regularity	% of patients who did their requested readings this week	Shows patients keep using the system
Control rate	% of patients with a 7-day home average below 135/85	The core health outcome
BP evolution	Group average systolic and diastolic over time	Shows the population improving
Alerts	Number of alerts per level per week	Shows what the system catches
Response time	Median time from red alert to doctor acknowledgement	Shows the loop works
Medication adherence	% of readings tagged "took my pill"	Explains uncontrolled cases
Screening	People screened, % flagged, % who then saw a doctor	Covers early detection

Each indicator is shown as a big number plus a small trend chart, filterable by period.

Architecture and stack

One Node backend receives readings from every channel, runs the alert and analytics logic, and pushes results to the doctor dashboard and back to the patient.

Every channel feeds one backend that alerts the doctor



system architecture · 3 input channels, 1 backend, 2 outputs

Whatever the channel, a reading is normalised into the same format before the alert engine sees it, so adding a channel later means adding one adapter.

Stack

Part	Technology
Patient app	React Native with Expo, local storage for offline, ML Kit for on-device OCR
Doctor dashboard	Next.js, Tailwind, TanStack Query, Recharts
Backend	Node.js with Express, Zod validation, JWT auth with roles
Database	PostgreSQL with Prisma
Real-time	Socket.IO for live dashboard updates

Part	Technology
SMS	Android phone with an SMS gateway app, forwarding to a webhook
Push	Expo push notifications
LLM	Hosted LLM API, restricted to the approved advice library
Deploy	Vercel for the dashboard, a small container for the backend

Data model

Table	Key fields
User	id, role (patient, doctor, pharmacist), phone, name, language
Patient	userId, doctorId, birthYear, sex, targetSys, targetDia, treatmentNames
Doctor	userId, availabilityHours, digestTime
Reading	id, patientId, sys, dia, pulse, takenAt, channel (app, sms, pharmacy), confirmed
ReadingTag	readingId, tag (pill_taken, pill_missed, salty, stress, poor_sleep)
Symptom	readingId, type (headache, dizziness, chest_pain, vision, confusion, weakness, speech, nosebleed)
Alert	id, patientId, level, reason, status (open, acknowledged, closed), createdAt, ackAt
Message	id, fromUserId, toUserId, body, createdAt
AdviceItem	id, situation, language, text, reviewedBy

Main API endpoints

- `POST /readings` from the app; `POST /sms/inbound` from the gateway; `POST /pharmacy/readings` from the form
- `GET /doctor/patients?sort=risk`, `GET /patients/:id/readings`, `GET /patients/:id/summary`
- `PATCH /alerts/:id` to acknowledge or close; `POST /messages`
- `GET /indicators?from=&to=`

Hackathon MVP

Build the full loop (reading, alert, doctor action) with a few features done well, rather than every screen half done.

Must have

- [ ] Patient app: onboarding, add reading (typed), context tags, symptoms, result screen, history chart
- [ ] Confirmation re-measure flow and the five alert levels
- [ ] SMS channel through the gateway phone, with automatic reply
- [ ] Doctor dashboard: triage list, alert inbox with explanations, patient page with chart
- [ ] Trend detection (moving average and slope) and the missed-pill pattern
- [ ] Indicators page
- [ ] Synthetic patients seeded for the demo

Nice to have

- [ ] OCR from a photo of the monitor
- [ ] LLM advice chat in Arabic, Darija and French
- [ ] Pre-consultation PDF report
- [ ] Adaptive reminders
- [ ] Pharmacy entry form and screening mode

Team split (4 people)

Person	Owns
1	Patient mobile app
2	Doctor dashboard and indicators
3	Backend, database, SMS gateway, alerts
4	Analytics (trends, patterns), seed data, advice library, pitch

Adjust to your team size; with 3 people, merge roles 3 and 4.

Demo scenario (about 3 minutes)

1. Show the doctor dashboard: 30 synthetic patients, sorted by risk; open one with a rising trend and read its explanation.
2. A patient enters 182/118 in the app; the app asks to re-measure; the second reading confirms; the patient ticks "headache".
3. The doctor's screen turns the patient red in real time; the doctor acknowledges and sends "Come to the clinic today".
4. Put a phone in airplane mode except SMS, send `TA 165 100 78`; the reading appears on the dashboard and the phone gets the reply.
5. End on the indicators page: control rate, regularity, response time.

Questions to expect from the jury

- Who validated the thresholds? Answer: guideline values, to be validated with a partner doctor.
- What if the doctor is unavailable? Answer: availability hours, escalation, and 14 for emergencies.
- How is patient data protected? Answer: role-based access, encryption in transit and at rest, each doctor sees only own patients.